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PALAEOMAGNETISM AND DEFORMATION OF THE MESOZOIC CONTINENTAL MARGIN IN SICILY *

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ABSTRACT

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The pre-orogenic morphology of the west Sicilian Mesozoic continental margin was characterised by platforms and basins elongated more or less parallel to the ancient junction between ocean and continent. The deformation of this continental margin during the Miocene gave rise to a number of thrust sheets which were transported southwards where they rest against the stable Iblean plateau. Eight thrust sheets have been sampled for palaeomagnetism in order to establish the amount of rotation, relative to Iblei, which occurred during emplacement. Clockwise rotations of large magnitude appear to have taken place, and these rotations are considered to be related to the emplacement of the Calabrian—Peloritani structure onto this continental margin.

INTRODUCTION

The geological structure of western Sicily is formed by a number of piled thrust sheets. The vergence of these imbricated units is to the south where they rest against the stable Iblean plateau (Fig. 1). The amount of tectonic transportation increases towards the north of the island where the tectonic units occupying the top of the thrust pile are cropping out. The timing of progressive deformation from north to south can be established stratigraphically because sedimentation was more or less continuous during the Neogene tectonic transport of the thrust sheets. A number of structural-stratigraphic units may be defined (Figs. 2 and 3). These units are geological bodies which display homogeneity in facies and tectonic behaviour, and are derived from pre-existing palaeogeographic units. The thrust planes which define the structural-stratigraphic units very often coincide with the zones of Mesozoic

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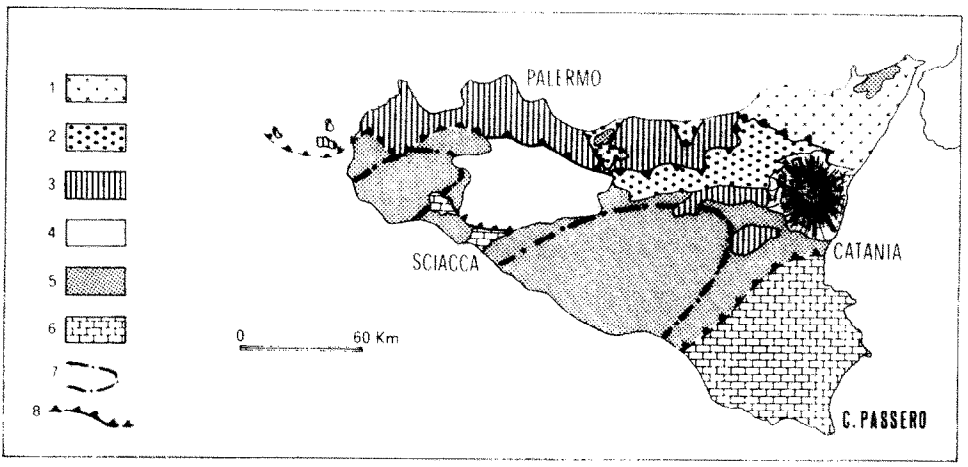
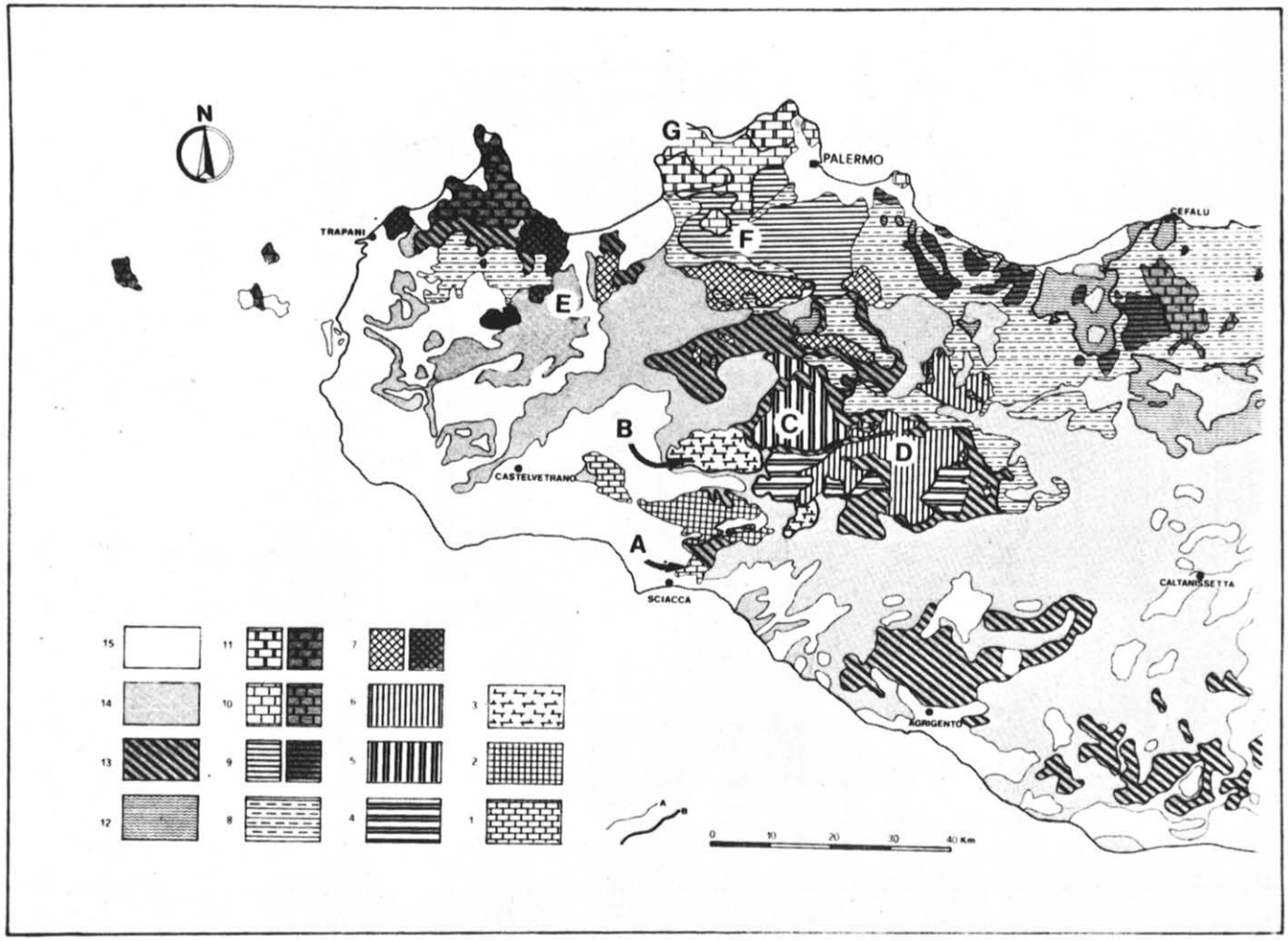


Fig. 1. Geological sketch of Sicily. *1* = Calabrian—Peloritani units; *2* = Sicilidi units; *3* = units derived from the Panormide, Imerese and Trapanese domains and deformed before Middle Tortonian; *4* = units derived from the Trapanese, Sicanian and Saccense domains and deformed after the Middle Tortonian; *5* = foredeep deposits; *6* = foreland; *7* = maximum subsidence in the foredeep, according to seismic data; *8* = main exposed and buried thrust fronts.

faulting at the boundaries of palaeogeographic zones, and the term structural-stratigraphic unit (D'Argenio and Scandone, 1970) is used in order to emphasise the close relationship between facies boundaries and tectonic discontinuities. Restoration of the pre-orogenic position of these structural-stratigraphic units can be performed by progressively withdrawing each unit with respect to that beneath. This type of palinspastic restoration of the west Sicilian structural-stratigraphic units along the N—S line between Palermo and Sciacca (Fig. 1) provides a schematic representation of the physiography of the sedimentary area before Neogene deformation (Fig. 4). The pre-orogenic morphology consisted of a number of carbonate platforms

Fig. 2. Map of structural-stratigraphic units in western Sicily (after Catalano and D'Argenio, 1978). *15* = Post-orogenic marine to continental deposits (Pliocene—Quaternary); *14—13* = Caltanissetta—Castelvetrano units: early Pliocene, Messinian and late Tortonian (*14*) and unconformably resting on Tortonian—Serravallian (*13*); *12* = Sicilidi units (s.l.): "argile varicolori"; *11* = Monte Gallo—Monte Palmeto unit; *10* = Cozzo di Lupo unit; *9* = Sagana—Belmonte Mezzagno unit; *8* = Piana degli Albanesi unit; *7* = Monte Kumeta and Rocca Busambra units; *6* = Monte Rose unit; *5* = Monte Barracu—Monte Colomba unit; *4* = Pizzo Mondello unit; *3* = Monte Genuardo unit; *2* = Pizzo—Telegrafo unit; *1* = Monte San Calogero unit. The shadowed symbols of blocks 7, 9, 10 and 11 indicate structural-stratigraphic units outside the Palermo—Sciacca section. Lines: *A* = stratigraphic and/or tectonic boundaries; *B* = thrust lines and their prolongation.

Palaeomagnetic sampling localities: *A* = Sciacca: Monte San Calogero unit; *B* = Sambuca: Monte Genuardo unit; *C* = Campofiorito: M. Barracu—M. Colomba unit; *D* = Filaga: Monte Rose unit; *E* = Monte Bonifato: Monte Kumeta unit; *F* = Sagana: Sagana—Belmonte Mezzagno unit; *G* = Terrasini/Cala Rossa: Cozzo di Lupo unit.


N
G
PALERMO
TRAPANI
CEFALU
F
E
B
C
D
CASTELVETRANO
A
SCIACCA
CALTANISSETTA
AGRIGENTO
15
14
13
12
11
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3
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1
A
B
0
10
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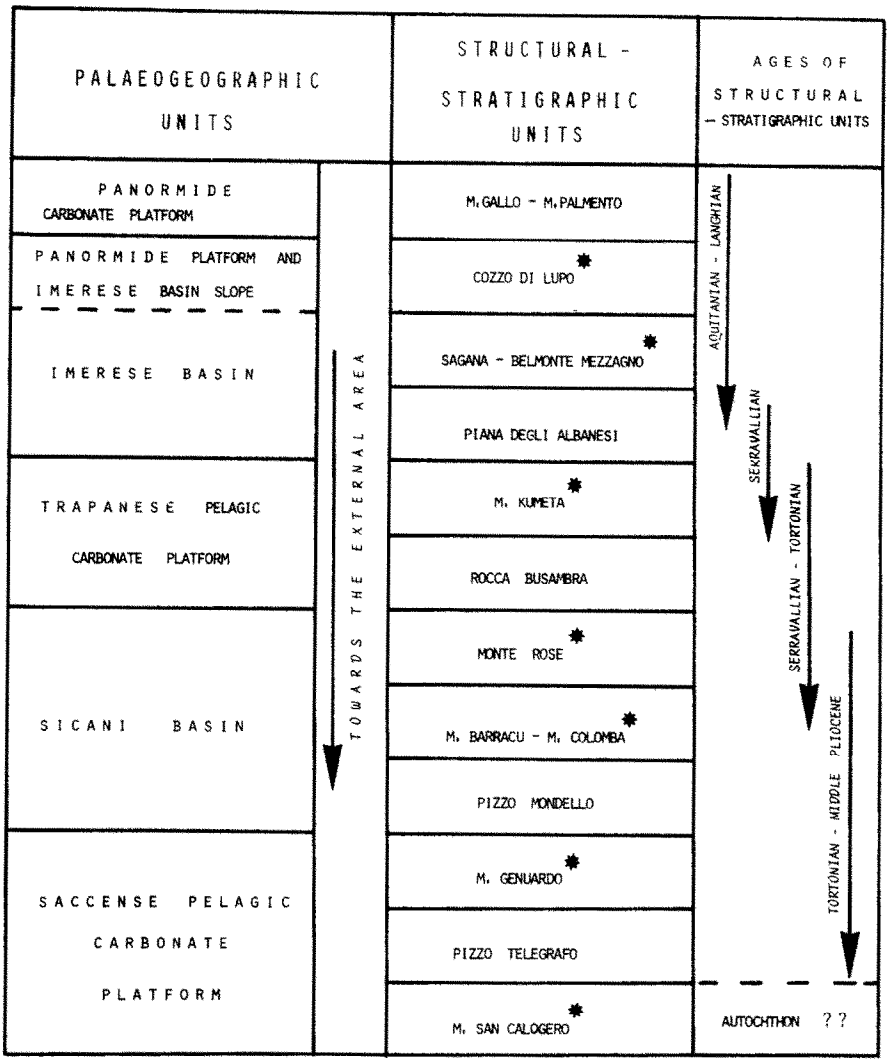


Fig. 3. The structural-stratigraphic units cropping out along the section between Palermo and Sciacca, and the palaeogeographic units from which the structural-stratigraphic units were derived. The asterisks denote the sampled structural-stratigraphic units.

and/or seamounts, with intermediate basins characterised by more siliceous carbonate sedimentation (see also Catalano et al., 1976; Catalano and D'Argenio, 1978). These platforms and basins were elongated more or less parallel to the oceanic—continental junction, the Panormide carbonate platform (Fig. 4) representing the most internal palaeogeographic unit of the continental margin. This continental margin is generally considered to have been part of the southern margin of the Mesozoic Tethys (Laubscher, 1971;

Fig. 4. The main lithologies characterising the palaeogeographic zones of the west Sicilian continental margin from Triassic to Lower Tertiary. The palaeogeographic section at the top of the figure depicts the inferred situation during the Upper Jurassic—Lower Cretaceous.

Bernoulli and Jenkyns, 1974; Scandone et al., 1974; D'Argenio et al., 1975; D'Argenio, 1976; Channell and Horváth, 1976) and these characteristic facies can be traced into the Apennines to the north, and into the African Maghrebids. We have studied palaeomagnetically several structural-stratigraphic units in order to ascertain their rotations during tectonic transport, relative to the stable Iblean foreland, and to estimate the orientation of the original palaeogeographic units (Fig. 4) from which the sampled structural-stratigraphic units were derived.

PALAEOMAGNETIC DATA

Previous palaeomagnetic work in Sicily has been concentrated in the Iblean zone of southeast Sicily (Fig. 1) (Schult, 1973; Barberi et al., 1974; Gregor et al., 1975). The Iblean region of Sicily is generally considered to be autochtonous, and it has been argued both on geological and palaeomagnetic grounds that this unit is an extension of the Saharan platform. The thrusted structural-stratigraphic and palaeogeographic units means that the palaeomag-deformed, therefore the magnetization directions and inferred rotations can be applied to each tectonic unit as a whole. The close relationship between structural-stratigraphic and palaeogeographic units means that the palaeomagnetic results from a number of such units can help to establish the original orientation of the continental margin on which they were rooted during Mesozoic and early Tertiary times. Upper Cretaceous and Paleocene magnetization directions from tectonic units derived from the Saccense, Sicani, Trapanese, Imerese and Panormide domains (Fig. 3) are compared with those from the Iblean autochthon.

Palaeomagnetic data from Iblei were derived from the Upper Cretaceous basalts of Capo Passero (Fig. 1). The basalts are overlain by Senonian rudistid limestones and Upper Cretaceous Scaglia-type pelagic calcilutites are known at their base from borehole data (Pieri, 1969). As the basalts have negative polarity the Upper Cretaceous magnetic stratigraphy (Lowrie and Alvarez, 1977; Channell et al., 1979) restricts their age to Campanian. We use a mean magnetic direction for these basalts (decl. = 165, incl. = —0.25), compiled from the data of Schult (1973), Barberi et al. (1974) and Gregor et al. (1975), to compare with the late Cretaceous directions from western Sicily.

A fine grained white/grey to red pelagic limestone, known as Scaglia, characterises the late Cretaceous and Palaeogene stratigraphy in Sicily (Fig. 4), and other parts of the Periadriatic continental margin (Bernoulli and Jenkyns, 1974; D'Argenio, 1976). The remanent magnetization of the Scaglia is usually dominated by magnetite of detrital origin (Lowrie and Alvarez, 1975; Channell et al., 1979), although in the red varieties haematite can be an important contributor, and can have a magnetization of diagenetic origin significantly post-dating the primary detrital magnetization (Channell, 1978). At most of the west Sicilian sampling localities, only the magnetite-

bearing white Scaglia was collected. However, in the case of the Cozzo di Lupo and Sagana—Belmonte Mezzagno structural-stratigraphic units (Fig. 3), the red Scaglia variety was collected because the white was found to be very weakly magnetized at these localities. As in certain previous studies of the Periadriatic Scaglia limestones, alternating field demagnetization techniques were sufficient to produce a stable characteristic magnetization for the white and pink Scaglia (Lowrie and Alvarez, 1975, 1977) but thermal techniques were necessary in the case of the red (Roggenthen and Napoleone, 1977; Channell et al., 1979).

The precise age of the Late Cretaceous Scaglia was not palaeontologically ascertained for all sampling sites. Any changes in magnetization direction during the time of deposition of this facies are insignificant compared to changes incurred by rotation during tectonic transport of the structural-stratigraphic units.

The Monte San Calogero structural-stratigraphic unit (Fig. 3) was sampled at a single locality close to Sciacca (Fig. 2). The site mean magnetization direction after alternating field demagnetization (Table I) is close to those from the Iblean zone of southeast Sicily (Fig. 5a).

The Monte Genuardo unit was derived from the deformation of the internal margin of the Saccenese platform (Fig. 3). Five sites were collected near the village of Sambuca (Fig. 2). The magnetization directions after demagnetization (Fig. 5b, Table I) deviate significantly from those obtained from the Monte San Calogero unit.

The Monte Barracu—Monte Colomba and the Monte Rose structural-stratigraphic units were both derived from the Sicani basin (Fig. 3). The Monte Barracu—Monte Colomba unit was sampled near the village of Campofiorito (Fig. 2) and the Monte Rose unit at the quarries near Filaga (Fig. 2). The site mean directions from these two localities (Fig. 5c and 5d, Table I) are similar.

The Monte Kumeta structural-stratigraphic unit was sampled at Monte Bonifato (Fig. 2). The site mean directions of 21 Maastrichtian—Paleocene sites (Fig. 6, Table II) are well grouped after alternating field demagnetization. The occurrence of normal and reversed polarities in more or less equal numbers is consistent with the reversal frequency in the Maastrichtian—Paleocene interval.

To the west of the Palermo mountains, along the coast between Terrasini and Cala Rossa (Fig. 2), 28 sites were collected from the white and red Upper Cretaceous—Paleocene Scaglia limestones belonging to the Cozzo di Lupo structural-stratigraphic unit. The magnetic properties of these varicoloured limestones have previously been discussed (Channell, 1978). The white variety of Scaglia at this locality, with intensities of the order of $5 \cdot 10^{-8}$ G, is usually too weakly magnetic for precise measurement with the Digico magnetometer (Molyneux, 1971). The study was concentrated on the red Scaglia limestones which have magnetic intensities about one order of magnitude greater than the adjacent white variety. One

TABLE I

Results of palaeomagneti measurements from Sicani and Saccense palaeographic units

Structural-stratigraphic unit	Site	No. of samples	N.R.M.			After 100 Oe			After 150 Oe			Mean
			decl.	incl.	α_{95}	decl.	incl.	α_{95}	decl.	incl.	α_{95}	
M. San Calogero	Sc 1	11	187.7	−8.3	34.6	342.7	32.6	7.5	354.7	34.7	6.9	
M. Genuardo	Sa 1	5	40.7	46.4	27.6	43.8	14.3	9.2	40.4	13.8	12.3	decl. = 47.6
	Sa 2	8	24.3	64.3	5.4	27.1	46.9	7.1	25.7	50.0	6.2	incl. = 40.5
	Sa 3	7	7.7	53.3	9.5	24.2	48.9	9.9	23.8	47.6	10.4	
	Sa 4	8	72.7	68.7	10.0	83.0	42.3	5.4	79.7	38.4	4.6	
	Sa 5	7	239.5	24.2	60.7	245.8	−41.6	9.5	244.5	−42.0	6.7	
M. Barracu— M. Colomba	Ca 1	9	92.4	55.5	4.4	127.9	45.1	7.2	134.2	43.4	10.1	decl. = 98.8
	Ca 2	8	301.3	−16.3	9.3	278.8	−31.6	4.8	275.7	−31.0	6.0	incl. = 44.4
	Ca 3	9	269.1	−75.2	24.6	272.7	−60.9	15.2	288.1	−56.1	19.6	
	Ca 4	9	97.1	45.9	8.3	103.2	40.2	4.4	96.8	39.5	3.3	
	Ca 5	9	236.5	−55.2	22.6	242.8	−41.3	22.0	243.6	−40.5	21.5	
M. Rose	Fi 1	6	294.8	−21.5	7.4	288.5	−48.4	5.4	286.2	−44.1	6.7	decl. = 100.3
	Fi 2	13	258.5	−17.1	7.8	268.5	−35.8	4.2	275.2	−35.2	5.8	incl. = 39.8

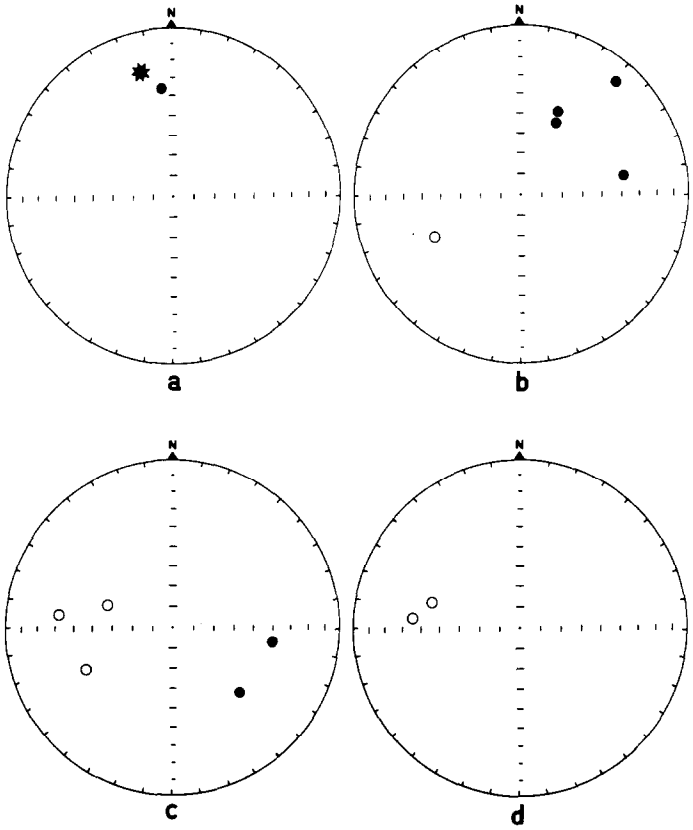


Fig. 5. Stereographic plots of site-mean magnetization directions after demagnetization in peak fields of 150 Oe. (a) Sciacca; (b) Sambuca, (c) Campofiorito; (d) Filaga. The closed and open symbols represent positive and negative inclinations, respectively. The star in (a) is the Upper Cretaceous mean direction derived from the Iblean zone of south-east Sicily.

pilot sample from each site was thermally demagnetized in stepwise fashion up to 600° C. Large directional changes occurred during thermal demagnetization but a stable magnetic direction could usually be isolated after thermal demagnetization at 400° C (Fig. 7). The characteristic remanence, isolated by thermal demagnetization, is carried by fine grained pigmentary haematite which was probably produced during diagenesis (Channell, 1978). The age of the magnetization post-dates the palaeontological age of the sediment. Within the Scaglia section at Terrasini, there are both tectonic and synsedimentary folds. Application of the fold test (Graham, 1949) to samples collected around the limbs of a tectonic and synsedimentary fold (Fig. 8) establishes that the characteristic magnetization was produced after synsedimentary folding and prior to tectonic folding.

The site mean directions (Fig. 9, Table III) after thermal cleaning are

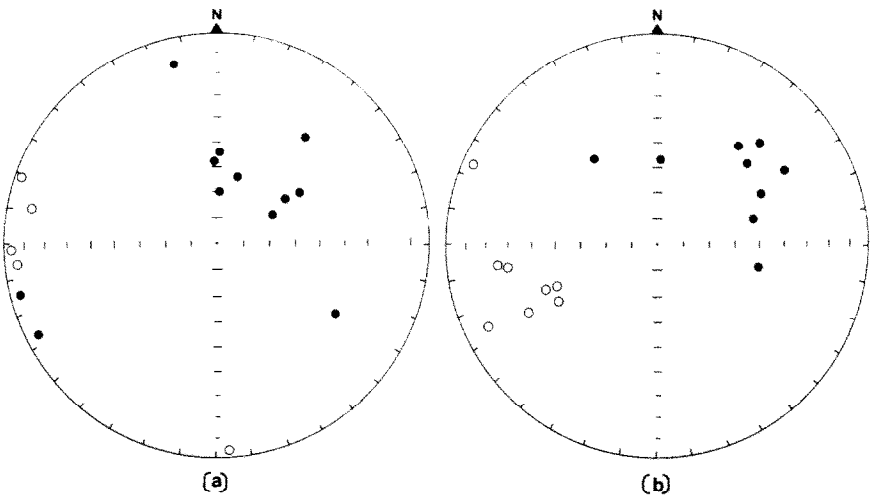


Fig. 6. Stereographic projections of the site mean directions from Monte Bonifato, (a) before and (b) after demagnetization. Closed and open symbols represent positive and negative inclinations, respectively.

TABLE II

Palaeomagnetic data from Monte Bonifato after alternating field demagnetization

Site	N	Decl.	Incl.	K	α_{95}
S 81	6	75.8	50.5	22.7	14.4
S 83	6	102.6	49.4	65.6	8.3
S 84	6	60.0	30.0	225.9	4.5
S 85	4	39.1	38.4	97.0	9.3
B 20	3	242.1	−31.6	64.8	15.4
B 22	5	248.5	−42.6	92.3	8.0
B 23	3	240.1	−45.1	15.6	22.6
B 25 *	4	102.6	27.6	2.8	68.3
B 26	5	48.6	42.3	21.6	16.8
B 27	3	323.2	48.3	59.3	16.1
B 28	5	247.2	−47.2	13.6	21.5
B 29	5	260.8	−28.0	55.7	10.3
B 30	5	262.1	−24.2	38.3	12.5
B 31	2	244.8	−11.1	499.4	11.2
B 32 *	2	250.4	62.8	58.7	33.2
B 33	4	293.4	−5.2	19.1	21.6
B 34	2	0.9	56.8	123.4	22.7
B 37 *	4	20.0	51.6	8.5	33.5
B 38 *	2	25.9	48.9	14.0	72.7
B 40	3	45.9	31.9	29.0	23.3
B 42	3	64.0	44.4	22.1	23.9
Mean	decl. = 63.4		N = 17	R = 15.11	
	incl. = 41.2		K = 8.4	α_{95} = 13.0	

* = rejected sites, (α_{95} > 25).

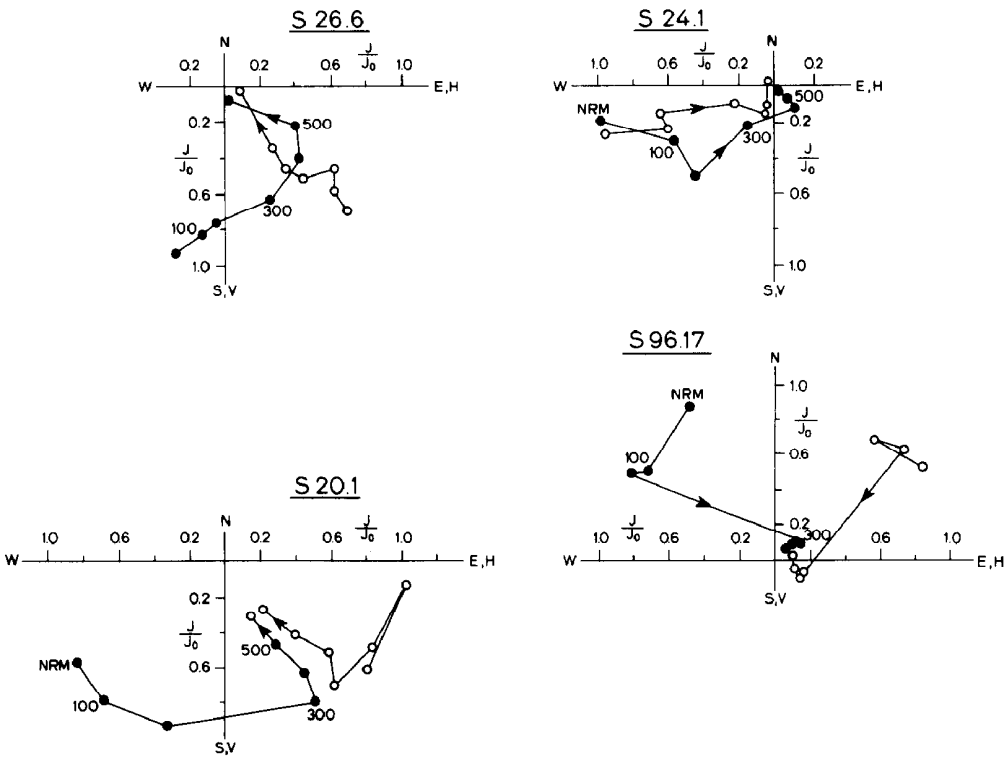


Fig. 7. Vector diagrams illustrating the changes in direction and intensity of remanence during thermal demagnetization for four samples from Terrasini. Solid and open symbols represent declination and inclination, respectively, and the temperature increments are shown in °C.

reasonably well grouped but there appears to be no systematic change in magnetic direction with age throughout the late Cretaceous—Paleocene sampling interval. This suggests that the growth of haematite did not occur at a constant rate during the diagenesis of the limestones. The paucity of negative polarities indicates that the critical phase of haematite growth usually occurred during period(s) of normal polarity.

The Sagana—Belmonte Mezzagno structural-stratigraphic unit was sampled near the village of Sagana (Fig. 2), where three sites were collected from the Upper Cretaceous Scaglia limestones. The characteristic magnetization direction obtained after thermal cleaning is similar for both red and white varieties. Two sites are of normal polarity and the other reversed (Fig. 9). The magnetic directions given by these three sites (Fig. 9, Table III) are close to those of the samples from Terrasini (Cozzo di Lupo structural-stratigraphic unit).

Schult (1976) has studied four palaeomagnetic sites from Cretaceous basalts at Custonaci, near Trapani (Fig. 2). These data come from a struc-

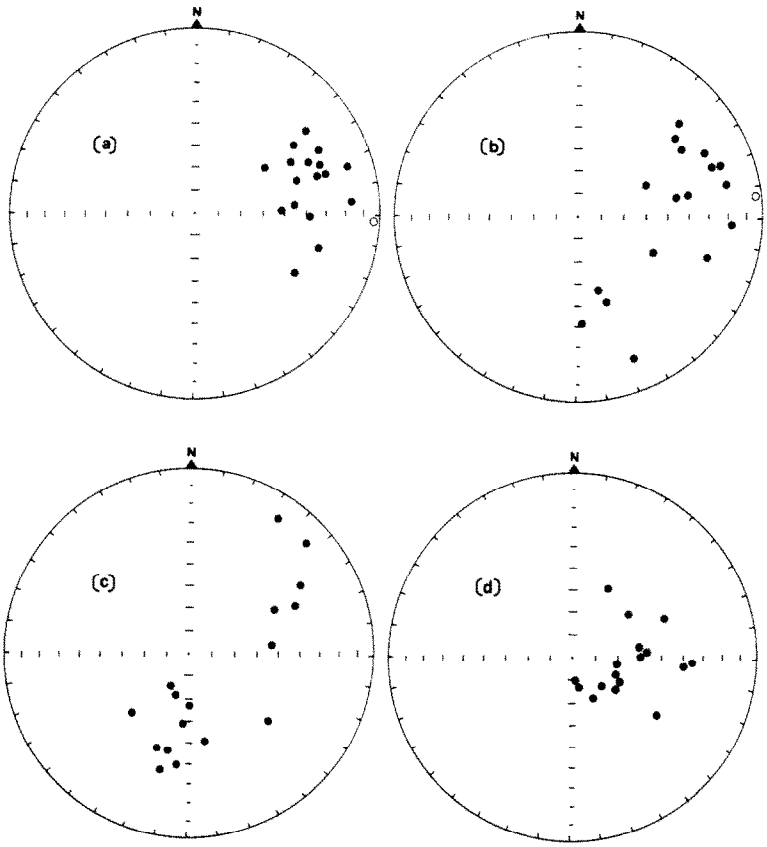


Fig. 8. Magnetically cleaned sample directions from a slump fold (a) before and (b) after bedding correction; and from a tectonic fold (c) before and (d) after bedding correction.

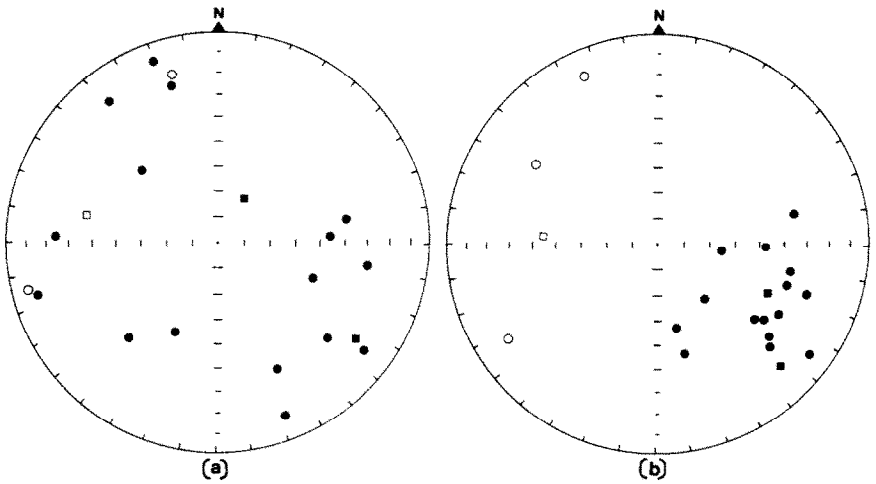


Fig. 9. Stereographic projections of the site mean directions from Terrasini (circles) and Sagana (squares) (a) before and (b) after thermal demagnetization. Closed and open symbols represent positive and negative inclinations, respectively.

TABLE III

Palaeomagnetic data from Terrasini and Sagana after thermal demagnetization

Site	N	Dec.	Inc.	K	α_{95}
S 3	6	109.5	25.4	68.5	8.1
S 4	6	108.9	35.1	82.3	7.4
S 6	6	303.9	−32.2	12.6	19.6
S 7	6	101.1	34.9	94.58	6.9
S 9	7	237.1	−17.3	64.3	7.6
S 10	8	167.7	55.8	10.86	17.6
S 11	8	90.4	47.3	22.42	12.0
S 17	2	127.5	39.5		
S 18	6	125.1	37.8	114.6	6.3
S 19	6	129.6	31.4	74.4	7.8
S 20	6	166.5	45.3	13.9	18.6
S 22	6	125.8	10.1	24.6	13.8
S 23	5	131.3	28.9	212.6	5.3
S 24	7	140.0	61.2	119.7	5.5
S 26	6	119.1	33.5	52.7	9.3
S 27 *	3	8.9	−64.4	2.34	90
S 28	6	335.8	−14.0	12.2	20.0
S 33 *	5	301.4	−54.4	2.81	56.2
S 96	18	77.0	33.2	16.3	8.8
S 97	17	96.4	64.8	15.4	9.4
S 56 **	7	134.6	17.2	136.9	5.2
S 58 **	7	113.1	41.5	81.5	6.7
S 61 **	7	274.4	−45.9	50.5	8.6
Overall mean		decl. = 124.0	N = 21		R = 19.28
		incl. = 38.1	K = 11.6		α_{95} = 9.7
Terrasini mean		decl. = 125.3	N = 18		R = 16.45
		incl. = 38.4	K = 11.0		α_{95} = 10.9
Sagana mean		decl. = 116.3	N = 3		R = 2.85
		incl. = 36.1	K = 13.0		α_{95} = 35.6

* = rejected sites ($\alpha_{95} > 25$); ** = sites from Sagana locality.

tural stratigraphic unit which was derived from the Panormide carbonate platforms but is probably not a simple extension of the Monte Gallo—Monte Palmeto unit of the Palermo Mountains (Figs. 2 and 3). The mean direction (decl. = 79, incl. = 29) of these four Cretaceous sites is well defined, and these data, derived from volcanic rocks, are compatible with the few samples which we have been able to measure from the very weakly magnetic platform carbonates which characterise the Panormide carbonate platform in the Palermo Mountains (Monte Palmeto, Monte Pellegrino). Schult's Jurassic sites collected from different structural-stratigraphic units derived from the deformation of the Imerese, Trapanese and Sicani palaeogeographic units are highly scattered. The scatter may reflect differential rotation of thrusted blocks within the sampled area, although the state of weathering of certain of the sampled volcanics may contribute to the dispersion.

DISCUSSION

The Monte San Calogero unit does not appear to have rotated relative to the Iblean unit, as the magnetization directions are similar (Fig. 5a). This tends to suggest that the Monte Calogero unit is a western equivalent of Iblei, and Iblei is generally accepted to be an autochthonous extension of the African platform (see Barberi et al., 1974).

The Monte Genuardo unit gives a mean declination (Table I) which differs from the Iblei Upper Cretaceous declination by about 60° implying a clockwise rotation of this order during the emplacement of this structural-stratigraphic unit. The Monte Barracu—Monte Colomba and Monte Rose s.s. units give mean declinations which suggest an even larger clockwise rotation (115°) relative to Iblei. The palaeomagnetic results from the Monte Kumeta structural-stratigraphic unit (Monte Bonifato) differ from the data of Iblei. The inclination is close to the Iblean result but the declination differs by 78°.

In the case of the white Scaglia, the primary contributor to the remanence is detrital magnetite, and the age of the magnetization can be associated with the stratigraphic age of the rock. The problems confronted at Terrasini of fixing the age of the magnetization do not occur. The mean magnetic directions may be interpreted in terms of clockwise rotation of these tectonic units relative to Iblei.

The mean direction from the Terrasini data (decl. = 125, incl. = 38), (Cozzo di Lupo structural-stratigraphic unit), is attributed to the growth of haematite during the diagenesis of the red Scaglia limestones. The fold test indicates that this magnetization was produced prior to tectonic folding and therefore before orogenic events affected the Sicilian continental margin in the Miocene. Also here, the mean magnetic direction is very different to that associated with the Upper Cretaceous basalts of Iblei. The inclination of the Terrasini data is only slightly higher than that from Iblei but the declination differs by 140°. This difference in declination may be interpreted in terms of large scale clockwise rotation of the Cozzo di Lupo structural-stratigraphic unit during its emplacement. The mean direction associated with the data from the Sagana—Belmonte Mezzagno (decl. = 116, incl. = 36) and Cozzo di Lupo (decl. = 125, incl. = 38) structural-stratigraphic units indicate that the two units have undergone comparable amounts of rotation.

Schult (1976) has implied that results from Cretaceous rocks at Custonaci and from Jurassic volcanics elsewhere in western Sicily are applicable to a west Sicilian "crustal block" which has rotated relative to Iblei, because an ancient plate boundary existed between Iblei and western Sicily. As mentioned above, Custonaci belongs to an allochthonous structural-stratigraphic unit derived from the deformation of the Panormide carbonate platform, and Iblei represents the autochtonous foreland against which this, and the other internal structural-stratigraphic units are thrusted. The degree of allochtony of the Sicilian part of this "Africa" continental margin increases north-

eastwards towards the impinging Calabrian—Peloritani structure (Fig. 1) which is made up of several piled tectonic units originally situated in more internal domains (Haccard et al., 1972; Alvarez et al., 1974; Amodio Morelli et al., 1976; Boccaletti et al., 1976). A plate boundary probably separated Calabria—Peloritani from this "African" continental margin but "western Sicily" and Iblei cannot be considered as having belonged to different plates. Nevertheless, the sense and order of magnitude (90°) of the rotations relative to Iblei implied by the Cretaceous data of Schult (1976) are compatible with the sense and order of magnitude of the rotations implied by our data.

We conclude that the tectonic units of the Sicilian continental margin have undergone large-scale clockwise rotation relative to the Iblean foreland. The data indicate also that the more internal tectonic units tend to have undergone more rotation, perhaps due to their greater distances of transport. Schult (1976) considered that the dextral shear which produced these rotations may have occurred as a result of dextral shear motion between the African and European plates. However, assuming that these tectonic rotations occurred during deformation of the continental margin, it is more likely that these rotations were a result of the emplacement of the Calabrian—Peloritani structures onto this continental margin. Although the onset of deformation of the west Sicilian continental margin has not yet been precisely established, it is thought that tectogenesis affected the Panormide carbonate platform from the early Middle Miocene (Wezel, 1970; Caflisch, 1966). The first overthrusting of the Calabrian—Peloritani massif onto the still rooted Apenninic palaeogeographic units (Campania—Lucania carbonate platform) appears to have taken place in the Aquitanian (Haccard et al., 1972). The rotations are an expression of the Neogene development of the arcuate shape of the Sicilian—Apenninic continental margin (Catalano et al., 1976). The "dextral shear" which would have affected the Sicilian portion of this continental margin during the emplacement of the Calabrian—Peloritani structure was probably the principle mechanism which produced the observed clockwise rotations.

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